COMP 3920 Database Systems 1

MySQL
Creating and Using Views

Introduction:
The goal of this document is to serve as a walkthrough creating some views in MySQL 8.

We are going to be using a sample database for a Classic Model Store.
For more information: https://github.com/hhorak/mysql-sample-db

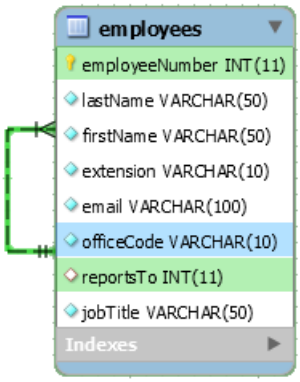
This Lab builds upon the Users, Roles and Privileges Lab where were created the HR, Sales, Management and Inventory Roles for the `classicmodels` database.

We will be creating 3 views:

1. A view to show employees and their bosses called **employeesAndBosses**.
2. A view to show the number of employees each boss has called **employeesPerBosses**.
3. A view to calculate product markups called **productMarkups**.

Step 1:
Using the mysql `root` account, create a view for all employees and their bosses.

The employees table in the `classicmodels` database has a foreign key back to itself. The `reportsTo` field is a foreign key to the `employeeNumber`.



We want to write a query that lists all employees and their bosses. We'll need a query that joins employees on itself. We also need to make sure our query includes ALL the employees (even if they don't have a boss – the owner of the company won't have a boss). Which query do we need?

Use this query (that has ALL employees) to create a view called employeesAndBosses.
MySQL Reference: https://dev.mysql.com/doc/refman/8.0/en/create-view.html

Run the following query to confirm your new view works:
`SELECT * FROM employeesAndBosses;`

Submit your screenshot:

01_employeesAndBosses_view.jpg

Submit

```
select * from employeesandbosses;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

employeeNumber	lastName	firstName	extension	email	officeCode	reportsTo	jobTitle	bossEmployeeNumber	bossLastName	bossFirstName
1002	Murphy	Diane	x5800	dmurphy@classicmodelcars.com	1	NULL	President	NULL	NULL	NULL
1056	Patterson	Mary	x4611	mpatterso@classicmodelcars.com	1	1002	VP Sales	1002	Murphy	Diane
1076	Firrelli	Jeff	x9273	jfirrelli@classicmodelcars.com	1	1002	VP Marketing	1002	Murphy	Diane

Step 2:

Create a view that counts the number of employees each boss has.

Now that we have a view with each employee with their boss - write a query that uses `employeesAndBosses` and counts the number of employees each boss has (grouped by the `bossEmployeeNumber`).

Include the following columns:

- `count(employeeNumber)`
- `bossEmployeeNumber`
- `bossFirstName`
- `bossLastName`

Run your query; the result should look like this:

count(employeenumber)	bossEmployeeNumber	bossFirstName	bossLastName
1	NULL	NULL	NULL
2	1002	Diane	Murphy

Create a view from the above query. Call your new view `employeesPerBoss`.

Execute this query:
```
SELECT * FROM employeesPerBoss;
```

Submit your screenshot:
02_employeesPerBoss_view.jpg

Submit

```
46 •  select * from employeesperboss;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

count(employeenumber)	bossEmployeeNumber	bossFirstName	bossLastName
1	NULL	NULL	NULL
2	1002	Diane	Murphy

Step 3:
We are going to create one more view to help those in sales calculate the markup on the products.

Markup is the amount added to the cost price of goods to cover overhead and profit.

Write a query that calculates the markup of items in the `products` table.

Include the following columns:

- `productCode`
- `productName`
- `buyprice`
- `msrp`
- `(msrp-buyprice)/buyprice*100 AS markup`
- `concat("+",round((msrp-buyprice)/buyprice*100,1),"%") AS markupFormatted`

There are 2 columns for markup. One that provides a numerical value for further calculations, the other is formatted nicely for display on the screen.

Run your query; the results should look like this:

productCode	productName	buyprice	msrp	markup	markupFormatted
S10_1678	1969 Harley Davidson Ultima...	48.81	95.70	96.066380	+96.1%

Create a view from the above query. Call your new view `productMarkups`.

Execute this query:

```
SELECT * FROM productMarkups
WHERE markup > 75;
```

Submit your screenshot:

03_productMarkups_view.jpg

Submit

```
59 • select *
60   from productMarkups
61   where markup > 75;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

productCode	productName	buyprice	msrp	markup	markupFormatted
S10_1678	1969 Harley Davidson Ultima...	48.81	95.70	96.066380	+96.1%